

CodeAlpha VIP Internship Program

Data Science Internship Report

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Internship Period: 20 July 2026 – 20 August 2026

**4-WEEK
VIRTUAL INTERNSHIP**

Internship Overview

Duration

20 July 2026 to 20 August 2026
A structured 4-week virtual internship in Data Science.

Internship Focus

Bridging theoretical Machine Learning concepts with hands-on work on structured tabular datasets.

Practical Tasks

1. Iris Flower Classification
2. Car Price Prediction with Machine Learning

- Worked through the complete data science workflow: preprocessing, feature engineering, model training, evaluation and visualization.
- Applied industry-standard Python libraries and statistical techniques to classification and regression problems.
- Completed task-based practical assignments and received a CodeAlpha Certificate of Completion.

Organization Profile — CodeAlpha

- Technical organization focused on skill development and hands-on virtual internships for engineering students.
- Registered under the Ministry of Micro, Small & Medium Enterprises (MSME) and the Ministry of Corporate Affairs (MCA), Government of India.
- Creates structured, project-oriented learning environments in Data Science, Artificial Intelligence and Full-Stack Software Engineering.
- Mission: “Your Dream Our Passion”.

Leadership

Ms. Swati Srivastava
Founder & CEO

Internship Approach

Project-driven learning with practical problem statements and hands-on implementation.

Core Domain

Data Science & Machine Learning

Problem Statement

Task 1 — Multi-Class Species Classification

Use petal and sepal measurements to classify iris specimens into Setosa, Versicolor and Virginica. The challenge is handling close morphological similarities and non-linear feature distributions.

Task 2 — Continuous Price Estimation

Build a regression system to predict automobile selling prices from heterogeneous factors such as brand, horsepower, mileage, vehicle age and fuel type while addressing non-linear interactions and multi-collinearity.

Project Objectives

EDA & Cleaning

Exploratory analysis and data cleansing using Pandas and NumPy.

Preprocessing

Outlier detection, missing-value handling and StandardScaler normalization.

Classification

Logistic Regression, SVC, Random Forest and k-NN for multi-class prediction.

Regression

Linear Regression, Ridge and Gradient Boosting for continuous price estimation.

Evaluation

Accuracy, F1-Score, Confusion Matrix, MAE, RMSE and R^2 Score.

Technologies & Tools Used

Programming

Python 3.10+

Data Manipulation

Pandas, NumPy

Machine Learning

Scikit-learn

Visualization

Matplotlib, Seaborn

Development

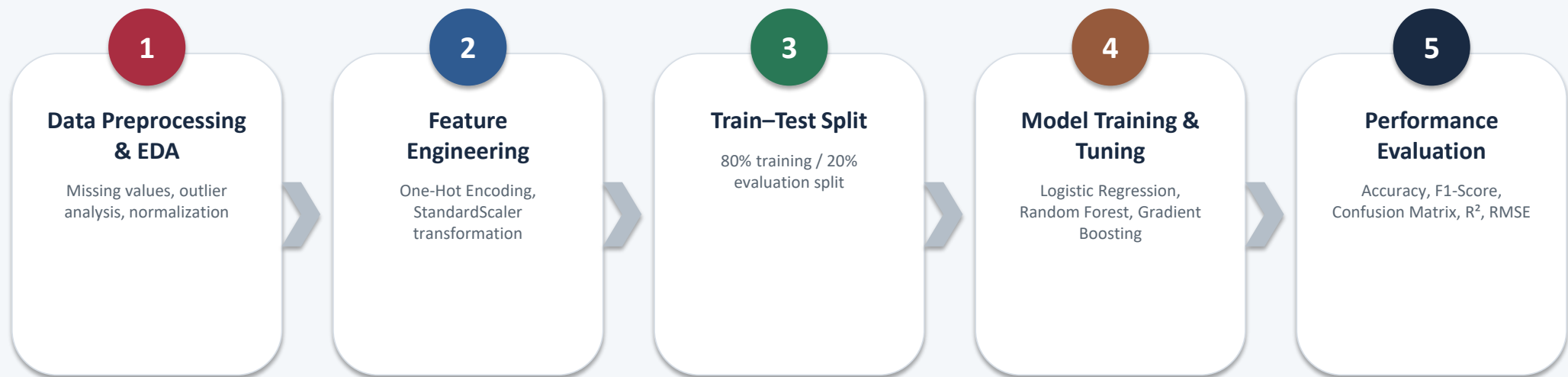
Jupyter Notebook, Visual Studio
Code

Version Control

Git, GitHub

Open-source Python ecosystem used across the complete workflow—from data ingestion to model evaluation.

End-to-End Data Science & ML Workflow



RAW INPUT DATA → ANALYSIS → TRANSFORMATION → MODELS → INSIGHTS

Task 1 — Iris Flower Classification

Dataset & Target

Four morphological features:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

Target: Setosa, Versicolor and Virginica

Methodology

1. Exploratory Data Analysis using scatter plots and pairplots.
2. Feature normalization using StandardScaler.
3. Training and evaluation of multiple classification algorithms.

Key Observation

Setosa was found to be linearly separable, while Versicolor and Virginica required more complex decision boundaries.

Task 1 — Model Performance

Model	Training Accuracy	Testing Accuracy	Macro F1
Logistic Regression	97.50%	96.67%	0.966
SVC (RBF)	99.16%	100.00%	1.000
Random Forest (n=100)	100.00%	96.67%	0.967
k-NN (k=5)	96.67%	96.67%	0.967

Best Model

Support Vector
Classifier (RBF)

100% Test Accuracy
Macro F1 = 1.000

Result Insight

The SVC achieved the
strongest reported test
performance among
the evaluated models.

Task 2 — Car Price Prediction

Brand / Make

Categorical reputation and market-tier factor

Horsepower & Engine Size

Primary numerical driver of performance and price class

Mileage & Fuel Economy

Fuel efficiency features with inverse pricing correlation

Vehicle Age / Model Year

Captures depreciation over operational lifespan

Methodology

One-Hot Encoding for nominal categorical features; log transformation of target price to mitigate skewness; comparison of Linear Regression, Ridge, Random Forest and Gradient Boosting.

Task 2 — Regression Performance

Regression Model	MAE (\$)	RMSE (\$)	R ² Score
Linear Regression (Baseline)	1,840.25	2,410.80	0.812
Ridge Regression	1,795.10	2,360.40	0.825
Random Forest Regressor	1,120.45	1,650.30	0.914
Gradient Boosting Regressor	980.20	1,420.15	0.938

Best Performer

Gradient Boosting
Regressor

$R^2 = 0.938$

RMSE = \$1,420.15

Fit Interpretation

Reported regression fit showed a tight residual distribution around the ideal 45° prediction axis.

Expected Outcomes & Skills Acquired

End-to-End Pipeline Proficiency

Clean, preprocess, scale and transform multi-type tabular data.

Classification Expertise

Train, tune and validate multi-class machine learning classifiers.

Regression & Valuation Modeling

Apply ensemble methods such as Gradient Boosting and Random Forest to continuous prediction.

Statistical Evaluation Rigor

Interpret confusion matrices, classification reports, MAE, RMSE and R^2 metrics.

Conclusion

Internship Summary

The internship provided practical exposure to the complete Data Science and Machine Learning workflow. Two complementary tasks demonstrated supervised learning across classification and regression: Iris species prediction achieved a reported 100% test accuracy with an RBF Support Vector Classifier, while car price prediction achieved the strongest reported performance with Gradient Boosting ($R^2 = 0.938$).

Thank You

Questions & Discussion

Selected References

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- Hastie, Tibshirani & Friedman — The Elements of Statistical Learning, 2nd Edition (2009)
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